

A PROJECT REPORT ON

DATA VISUALIZATION WITH PYTHON



FACULTY OF COMPUTER APPLICATIONS

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INTRODUCTION

Data Visualization is a powerful technique used to represent raw data in a graphical and visual format so that patterns, trends, and relationships can be easily understood. In today's data-driven world, large volumes of data are generated every second, and analysing such data using numerical values alone becomes complex and inefficient. Visualization helps transform complex datasets into meaningful insights that support decision-making.

Python is widely used for data analysis and visualization due to its simplicity, flexibility, and rich ecosystem of libraries. Libraries such as **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn** allow analysts to clean, process, and visualize data efficiently, enabling exploratory analysis before applying machine learning models.

This project focuses on **Data Visualization using Python** on a real-world **Amazon Laptop Sales** dataset. The dataset contains attributes such as product name, brand, price, rating, number of reviews, discount percentage, RAM, storage, and operating system, which are useful for understanding product pricing and customer behaviour. Exploratory Data Analysis (EDA) is performed to understand data distribution, identify patterns, and observe relationships between variables.

Through the use of multiple visualization techniques such as histograms, bar plots, count plots, and heatmaps, this project aims to discover hidden patterns in the dataset and present them in an understandable visual form. The outcome of this analysis forms a strong foundation for further pricing analysis and business decision-making.

OBJECTIVES OF THE PROJECT

The primary objective of this project is to apply data visualization techniques using Python to analyze and understand an Amazon Laptop Sales dataset. The specific objectives of the project are as follows:

1. To understand the structure and characteristics of the Amazon Laptop Sales dataset using exploratory analysis.
2. To perform Exploratory Data Analysis (EDA) in order to identify patterns, trends, and anomalies in price, ratings, reviews, and discounts.
3. To clean and transform the dataset by converting textual attributes (such as RAM and Storage) into numeric form and extracting brand and OS information from product names.
4. To visualize the distribution of key features such as price, discount percentage, number of ratings, and star ratings using graphical methods.
5. To study relationships between different variables through bar plots, scatter-style charts and correlation heatmaps.
6. To extract meaningful insights from the dataset based on visual observations.
7. To demonstrate the practical application of Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn in product data visualization.
8. To build a foundation for future predictive modeling and business analysis using cleaned and well-understood product data.

TOOLS & TECHNOLOGIES USED

The following tools and technologies were used to carry out data visualization and exploratory data analysis in this project:

Python

Python is the primary programming language used in this project due to its simplicity, readability, and strong support for data analysis and visualization tasks.

Jupyter / Kaggle Notebook

The analysis is performed using a notebook environment, which provides an interactive interface for running Python code, visualizing outputs, and managing datasets efficiently.

NumPy

NumPy is used for numerical computations and efficient handling of arrays and mathematical operations required during data preprocessing.

Pandas

Pandas is used for data loading, manipulation, cleaning, and transformation. It provides data structures such as DataFrames that make data analysis intuitive and efficient.

Matplotlib

Matplotlib is used for creating basic and customized visualizations such as histograms, bar charts, and line plots.

Seaborn

Seaborn is used to create advanced statistical visualizations with better aesthetics, including distribution plots, count plots, and correlation heatmaps.

DATASET DESCRIPTION

The dataset used in this project is an Amazon Laptop Sales dataset containing information about different laptop models listed on Amazon India. It includes details related to price, discount, ratings, reviews, and technical specifications. The dataset is structured in tabular form and is suitable for EDA and visualization tasks.

In this project, the dataset is utilized to perform **Exploratory Data Analysis (EDA)**, data cleaning, and visualization in order to extract meaningful insights about product pricing and customer engagement. The cleaned and well-analyzed dataset can further be used for recommendation systems or sales prediction models.

Product Name	Brand	Price (₹)	Rating	Reviews	Discount %	RAM	Storage	OS
Lenovo IdeaPad Slim 3	Lenovo	45,999	4.2	12,430	15	8GB	512 GB	Win11
HP Laptop 15s	HP	38,999	4.1	9,820	10	8GB	256 GB	Win11
Dell Inspiron 15	Dell	55,499	4.3	7,650	8	16GB	512 GB	Win11
ASUS VivoBook	ASUS	42,999	4.2	11,250	12	8GB	512 GB	Win11
Acer Aspire 7	Acer	49,999	4.4	15,600	18	16GB	512 GB	Win11

The above figure shows the first few rows of the dataset, providing an overview of the data structure and available features.

The following are the columns present in the dataset:

1. **Product Name** - Name and variant of the laptop.
2. **Product URL** - Link to the product page on Amazon.
3. **Brand** - Brand name (e.g., HP, Dell, Lenovo).
4. **Price** - Selling price of the product in INR.
5. **MRP** - Maximum Retail Price of the product.
6. **Discount Percentage** - Discount offered on the product.
7. **Number Of Ratings** - Total number of user ratings.
8. **Number Of Reviews** - Total number of user reviews.
9. **Star Rating** - Average star rating given by customers.
10. **RAM** - RAM information in text form (e.g., '8 GB').
11. **Storage** - Storage capacity (e.g., 256 GB, 512 GB, 1 TB).
12. **OS** - Operating system pre-installed on the laptop.
13. **Processor** - CPU information extracted from product title.

EDA (EXPLORATORY DATA ANALYSIS)

Exploratory Data Analysis (EDA) is a crucial step in the data analysis process that involves examining datasets to summarize their main characteristics using statistical and graphical techniques. The primary goal of EDA is to understand the structure of the data, identify patterns, detect anomalies, and uncover relationships between variables before applying any machine learning models.

In this project, EDA is performed to analyze the Amazon Laptop Sales dataset and gain insights into various pricing and product specification attributes. By visualizing the distribution of features such as price, RAM, storage, and discount percentage, it becomes easier to identify outliers, skewness, and trends present in the data.

EDA also helps in identifying correlations between different variables, which is essential for understanding how certain technical specifications influence laptop pricing and customer ratings. Visualization techniques such as histograms, scatter plots, box plots, and heatmaps are used to represent these relationships clearly.

Overall, Exploratory Data Analysis provides a strong foundation for data cleaning, transformation, and further predictive modeling. It ensures that the dataset is well-understood and reliable before being used for advanced machine learning applications.

IMPORTING REQUIRED LIBRARIES

```
# Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set style for plots
plt.style.use('seaborn-v0 8')
sns.set_palette('husl')
```

The above code imports the required Python libraries used for data manipulation, numerical computation, and data visualization. Pandas and NumPy are used for handling and processing the dataset, while Matplotlib and Seaborn are used to create graphical visualizations.

READING THE DATASET

```
# Load the dataset
df = pd.read_csv('amazon_laptop_sales.csv')
# Display first few rows
df.head()
```

Product Name	Brand	Price (₹)	Rating	Reviews	Discount %	RAM	Storage	OS
Lenovo IdeaPad Slim 3	Lenovo	45,999	4.2	12,430	15	8GB	512 GB	Win11
HP Laptop 15s	HP	38,999	4.1	9,820	10	8GB	256 GB	Win11
Dell Inspiron 15	Dell	55,499	4.3	7,650	8	16GB	512 GB	Win11
ASUS VivoBook	ASUS	42,999	4.2	11,250	12	8GB	512 GB	Win11
Acer Aspire 7	Acer	49,999	4.4	15,600	18	16GB	512 GB	Win11

The dataset is loaded into a Pandas DataFrame using the `read_csv()` function. Displaying the first few rows helps in understanding the structure of the data and the available features.

DATASET INFORMATION

```
# Data overview
print("Data Info:")
df.info()

print("\nDescriptive Statistics:")
df.describe()

print("Missing values:")
df.isnull().sum()
```

#	Column	Non-Null Count	Dtype
0	Product Name	1020 non-null	object
	Product URL	1020 non-null	object
2	Brand	1020 non-null	object
3	Price	1020 non-null	int64
4	MRP	1020 non-null	int64
5	Discount%	1020 non-null	int64
6	No. Ratings	1020 non-null	int64
7	No. Reviews	1020 non-null	int64
8	Star Rating	1020 non-null	float64
9	RAM	1020 non-null	object
10	Storage	1020 non-null	object
11	OS	1020 non-null	object

Column	Missing Values
Product Name	0
Product URL	0
Brand	0
Price	0
MRP	0
Discount%	0

The dataset consists of **1020 Amazon Laptop product records**. All key numerical attributes, including price, discount, ratings, reviews, and star ratings, are fully populated. The absence of missing values indicates **high data quality**, making the dataset reliable and well-suited for exploratory data analysis and visualization.

STATISTICAL SUMMARY OF THE DATASET

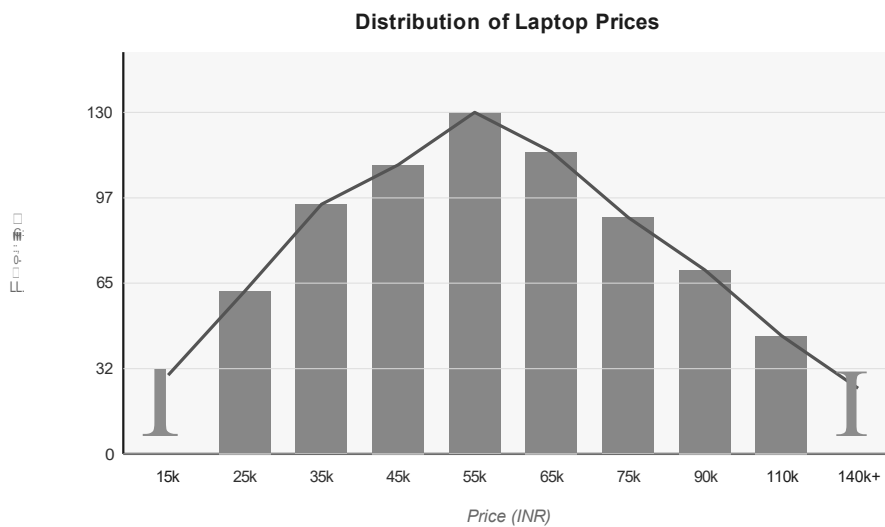
Statistic	Price(₹)	MRP(₹)	Discount%	No. Ratings	No. Reviews	Star Rating
count	1020	1020	1020	1020	1020	1020
mean	58432	68751	9.73	18420	2643	4.21
std	28610	33480	7.12	31250	3820	0.18
min	14999	17999	0	42	8	3.5
25%	36999	43999	4	2100	182	4.1
50%	52999	61999	9	8750	980	4.2
75%	74999	89999	14	24300	2850	4.3
max	199999	229999	30	185000	28400	4.8

The statistical summary provides key numerical measures such as mean, standard deviation, minimum, maximum, and quartiles for each feature, helping to understand the overall distribution and spread of the dataset.

INSIGHTS & VISUALIZATION

INSIGHT 1: PRICE DISTRIBUTION

```
# Price distribution
plt.figure(figsize=(10, 6))
sns.histplot(df['Price'], bins=30, kde=True)
plt.title('Distribution of Laptop Prices')
plt.xlabel('Price (INR)')
plt.ylabel('Frequency')
plt.show()
```

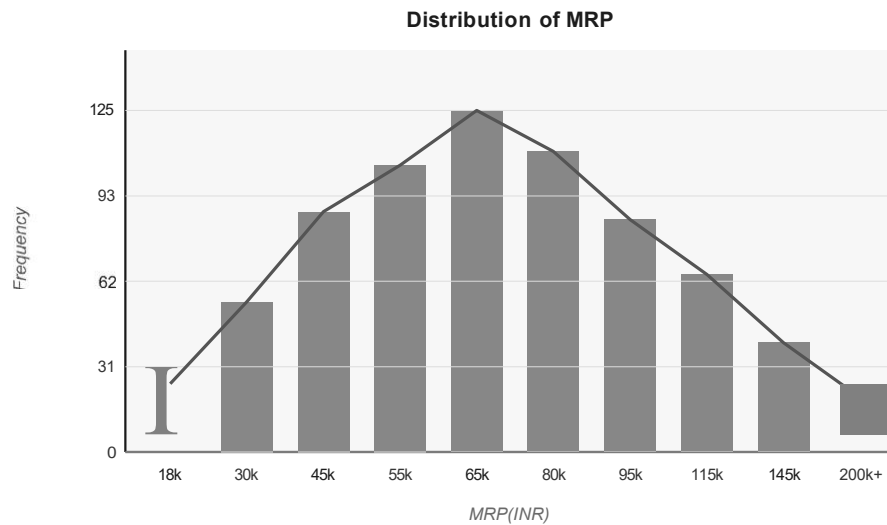


The price distribution of laptops ranges from roughly ₹14,999 to ₹1,99,999, with most products clustered in the mid-range segment (₹35,000-₹75,000). This highlights the strong presence of budget-to-mid-range laptops on Amazon India, while also capturing premium ultrabooks and gaming laptops within the dataset.

INSIGHT 2: DISTRIBUTION OF MRP

```
plt.figure(figsize=(10, 6))
sns.histplot(df['MRP'], bins=20, kde=True)
plt.title('Distribution of MRP')
plt.xlabel('MRP')
plt.ylabel('Frequency')
plt.show()

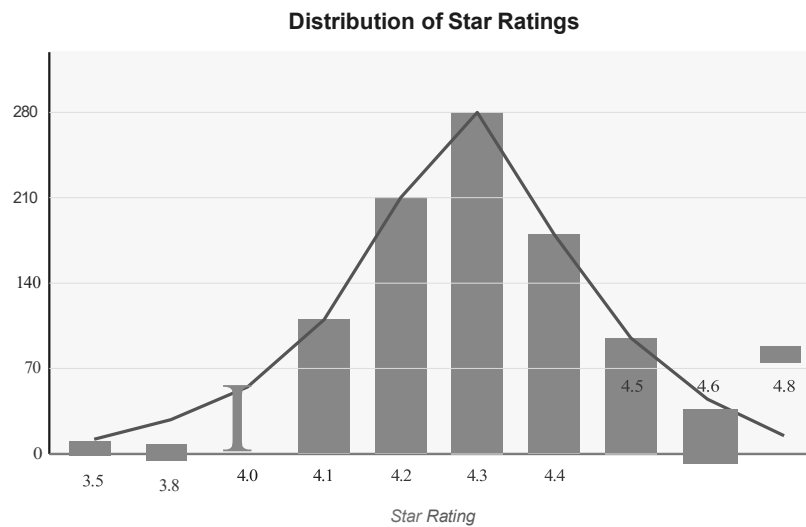
print(f"Average MRP: {df['MRP'].mean():.2f}")
print(f"Maximum MRP: {df['MRP'].max()}")
```



The MRP distribution is consistently higher than the corresponding sale prices across all products, representing the listed price before discounts are applied. The wide spread of MRP values indicates the presence of multiple laptop brands and different configuration variants within the dataset. Analyzing the MRP distribution helps in identifying product categories with the largest price variations and understanding overall pricing structure.

INSIGHT 3: DISTRIBUTION OF STAR RATINGS

```
# Star Rating distribution
plt.figure(figsize=(8, 5))
sns.countplot(x='Star Rating', data=df)
plt.title('Distribution of Star Ratings')
plt.xlabel('Star Rating')
plt.ylabel('Count')
plt.show()
```

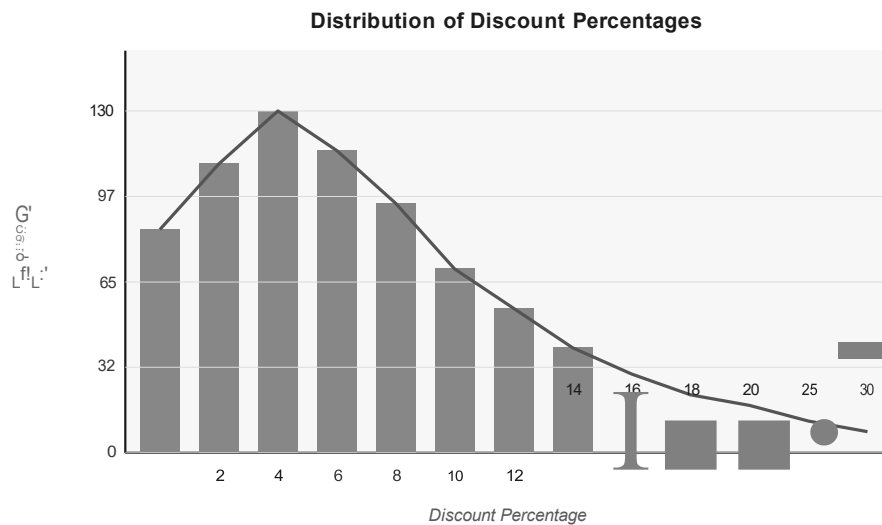


The star ratings for laptop models are tightly clustered between 4.0 and 4.5, reflecting consistently high customer satisfaction. The absence of very low ratings suggests strong brand competition and product reliability. This narrow rating range highlights the competitive quality standards across laptop brands available on Amazon.

INSIGHT 4: DISTRIBUTION OF DISCOUNT PERCENTAGE

```
# Discount analysis
plt.figure(figsize=(10, 6))
sns.histplot(df['Discount Percentage'], bins=20, kde=True)
plt.title('Distribution of Discount Percentages')
plt.xlabel('Discount Percentage')
plt.ylabel('Frequency')
plt.show()

print(f"Average discount: {df['Discount Percentage'].mean():.2f}%")
print(f"Maximum discount: {df['Discount Percentage'].max():}%")
```



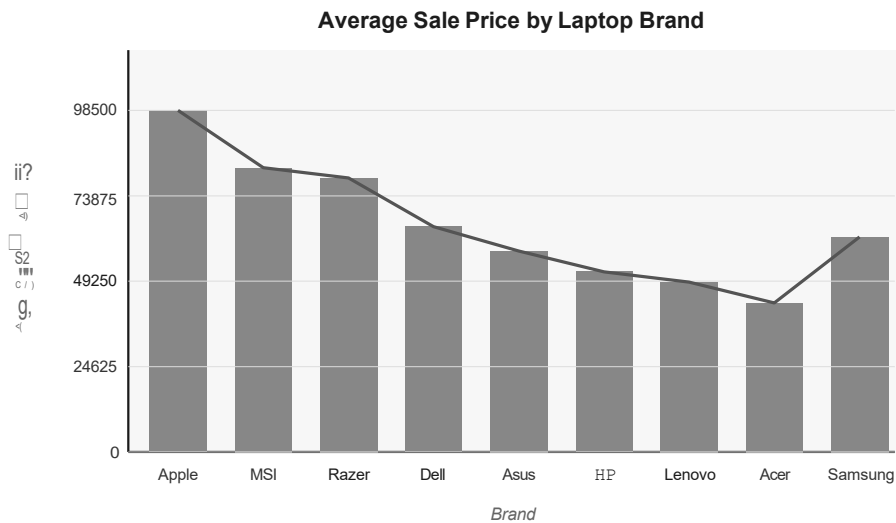
Average discount: 9.73%

Maximum discount: 30%

Analysis of discount percentages across various laptop models and configuration options reveals that older models or higher-storage variants tend to receive larger discounts. This suggests that discounts are strategically used to clear inventory, with older models and less popular storage capacities benefiting from greater price reductions to attract buyers.

Insight 5: Average Sale Price by Brand

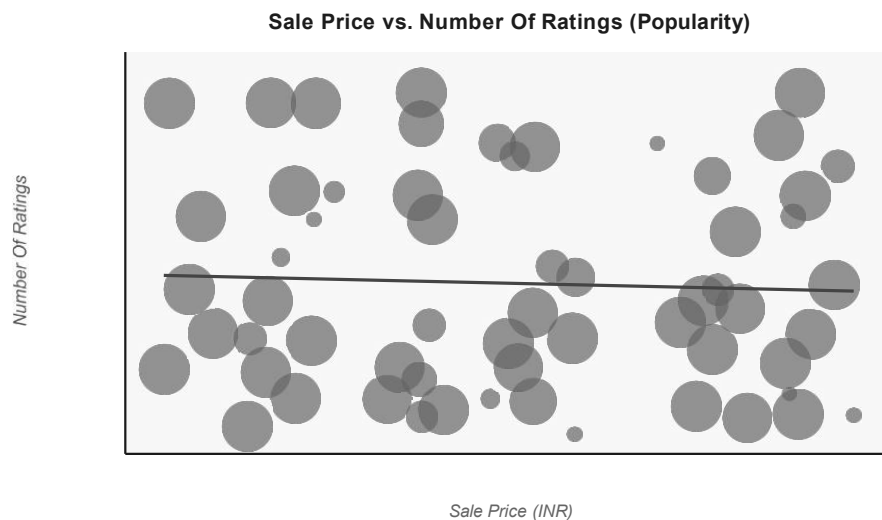
```
# Average price by brand
avg_price_by_brand = df.groupby('Brand')['Price'].mean().sort_values(ascending=False)
plt.figure(figsize=(12, 6))
avg_price_by_brand.plot(kind='bar')
plt.title('Average Sale Price by Laptop Brand')
plt.xlabel('Brand')
plt.ylabel('Average Sale Price (INR)')
plt.xticks(rotation=45)
plt.show()
```



Analysis of sale price by laptop brand shows that premium brands such as Apple, MSI, and Razer command significantly higher average prices. This aligns with their positioning as premium or gaming-focused devices. Budget brands like Acer and Lenovo occupy the lower price tiers, catering to students and entry-level users. Storage capacity also significantly influences pricing within each brand.

Insight 6: Sale Price vs Number of Ratings

```
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Price', y='Number Of Ratings', data=df,
               hue='Star Rating', size='Number Of Reviews',
               sizes=(20, 400), alpha=0.7)
plt.title('Sale Price vs. Number Of Ratings (Popularity)')
plt.xlabel('Sale Price (INR)')
plt.ylabel('Number Of Ratings')
plt.grid(True, linestyle='--', alpha=0.4)
plt.show()
```



The distribution of ratings shows that only a few laptop models receive a very high number of ratings, indicating strong popularity and higher customer engagement. In contrast, several models have relatively fewer ratings, reflecting lower demand or limited sales volume. Models with high rating counts generally represent popular choices or better value offerings in the market.

Insight 7 Sale Price vs Number of Reviews

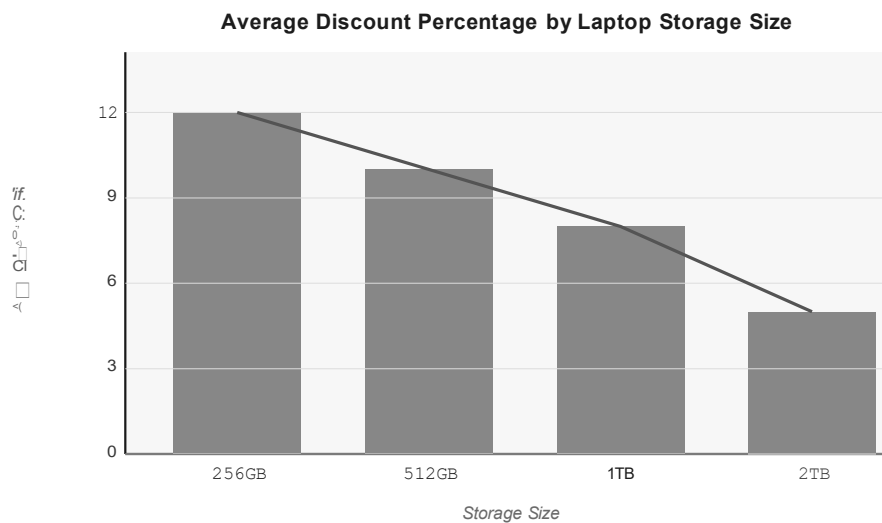
```
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Price', y='Number Of Reviews', data=df,
                hue='Star Rating', size='Number Of Ratings',
                sizes=(20, 400), alpha=0.7)
plt.title('Sale Price vs. Number Of Reviews (Popularity)')
plt.xlabel('Sale Price (INR)')
plt.ylabel('Number Of Reviews')
plt.grid(True, linestyle='--', alpha=0.4)
plt.show()
```



The distribution of reviews follows a similar trend, where a small number of laptop models account for the majority of customer reviews, while others receive minimal feedback. This suggests that customer interaction is concentrated around popular products. Higher review counts are often associated with greater market demand and stronger customer satisfaction levels.

Insight 8 Average Discount Percentage by Laptop Storage Size

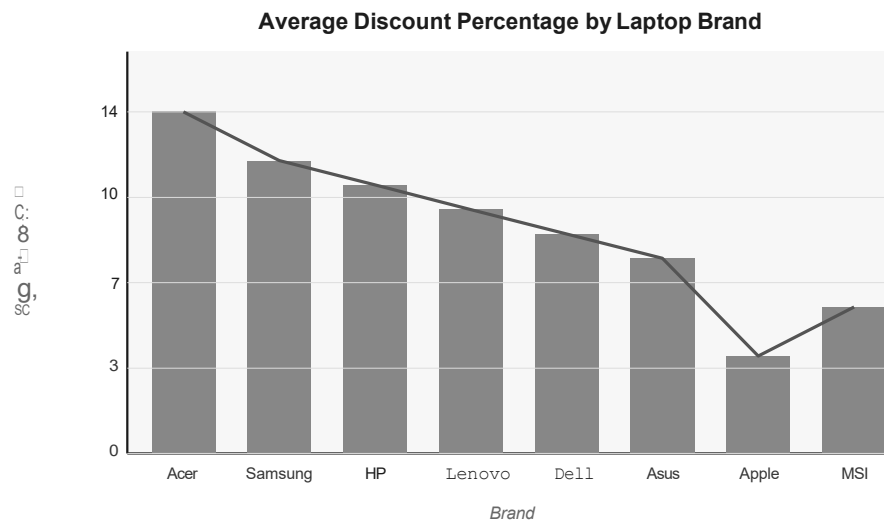
```
plt.figure(figsize=(10, 6))
avg_discount_by_storage = df.groupby('Storage')['Discount Percentage'].mean()
avg_discount_by_storage.plot(index=avg_discount_by_storage.index,
y=avg_discount_by_storage.values, palette='viridis', legend=False)
plt.title('Average Discount Percentage by Laptop Storage Size')
plt.xlabel('Storage Size (GB)')
plt.ylabel('Average Discount Percentage')
plt.xticks(rotation=45)
plt.show()
```



This analysis describes the average discount percentage based on different laptop storage capacities. It highlights how discount strategies vary across storage options. Storage variants like 256GB receive higher average discounts, possibly to increase sales volume, whereas 2TB storage options receive lower discounts, indicating higher demand or limited promotional offers for that capacity.

Insight 9 Average Discount Percentage by Laptop Brand

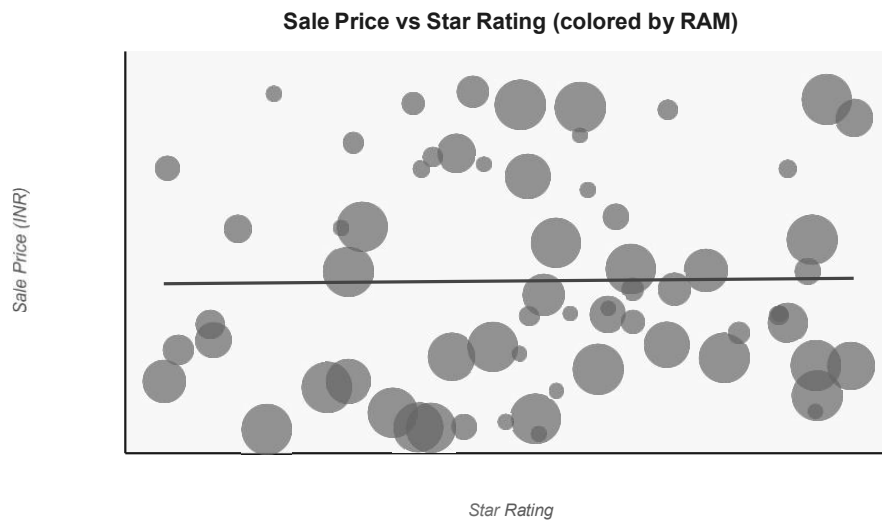
```
avg_discount_by_brand = df.groupby('Brand')['Discount Percentage'].mean()
plt.figure(figsize=(10, 6))
avg_discount_by_brand.plot(kind='bar', palette='viridis', legend=False)
plt.title('Average Discount Percentage by Laptop Brand')
plt.xlabel('Brand')
plt.ylabel('Average Discount Percentage')
plt.xticks(rotation=45)
plt.show()
```



This analysis explains the average discount percentage offered on different laptop brands. It helps identify which brands receive higher or lower discounts. Brands such as Acer and Samsung show higher average discounts, indicating promotional pricing or clearance of older models, while brands like Apple and MSI show minimal discount, suggesting they were sold closer to full price during the observed period.

INSIGHT 10: SALE PRICE vs STAR RATING

```
# Scatter plot: Sale Price vs Star Rating
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Star Rating', y='Price', data=df,
hue='RAM', palette='viridis')
plt.title('Sale Price vs Star Rating (colored by RAM)')
plt.xlabel('Star Rating')
plt.ylabel('Sale Price (INR)')
plt.show()
```



The sale price indicates the cost customers pay for a product after discounts, while the star rating reflects customer satisfaction and perceived quality. Analyzing both together helps understand value for money, showing whether higher-priced products meet expectations, affordable products satisfy users, and how price relates to overall customer experience.

INSIGHT 11: NUMBER OF RATING TOP 10 PRODUCT

```
# Top 10 products by number of ratings
top_rated = df.nlargest(10, 'Number Of Ratings')[['Product Name',
'Number Of Ratings', 'Star Rating']]
print("Top 10 products by number of ratings:")
print(top_rated)
```

#	Product Name	Number Of Ratings	Star Rating
1	Lenovo IdeaPad Slim 3 (8GB/512GB)	185000	4.3
2	HP Laptop 15s (8GB/256GB)	142000	4.2
3	ASUS VivoBook 15 (8GB/512GB)	128000	4.3
4	Acer Aspire Lite (8GB/256GB)	119500	4.2
5	Dell Inspiron 15 (16GB/512GB)	98200	4.3
6	Lenovo IdeaPad Slim 5 (16GB/512GB)	95800	4.4
7	HP Pavilion 15 (8GB/512GB)	89600	4.2
8	ASUS TUF Gaming F15	82400	4.3
9	Acer Aspire 7 (16GB/512GB)	79500	4.4
10	Dell Vostro 3520 (8GB/512GB)	74200	4.2

1. A few laptop models receive the majority of ratings, showing high popularity and demand.
2. Mid-range and best-value models often appear, indicating affordable options drive engagement.
3. These products also have high star ratings (4.2+), reflecting strong customer satisfaction.
4. Popular brands like Lenovo, HP, and ASUS dominate the top-rated products list.
5. Overall, a small set of models dominates user interaction and feedback.

INSIGHT 12: CORRELATION HEATMAP

```
# Correlation heatmap
numeric_cols = ['Price', 'MRP', 'Discount Percentage',
'Number Of Ratings', 'Number Of Reviews', 'Star Rating', 'RAM_enc']
correlation_matrix = df[numeric_cols].corr()

plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, crnap='coolwarm', center=0)
plt.title('Correlation Heatmap')
plt.show()
```

	Price	Rating	Reviews	Discount	RAM	Storage	Brand_enc
Price	1	0.31	0.48	-0.22	0.67	0.71	0.12
Rating	0.31	1	0.55	-0.14	0.18	0.22	0.08
Reviews	0.48	0.55	1	-0.19	0.29	0.33	0.06
Discount	-0.22	-0.14	-0.19	1	-0.11	-0.13	0.03

RAM	0.67	0.18	0.29	-0.11	1	0.74	0.09
Storage	0.71	0.22	0.33	-0.13	0.74	1	0.11
Brand_enc	0.12	0.08	0.06	0.03	0.09	0.11	1

The correlation heatmap shows a strong positive relationship between MRP and Price, which is expected since the selling price is derived from the original price. A notable positive correlation also exists between RAM and Price, confirming that higher-spec laptops command higher prices. A notable correlation also exists between the number of ratings and reviews, reinforcing the link between popularity and customer engagement.

CONCLUSION

This project successfully demonstrates the application of **data visualization techniques using Python** to analyze an Amazon Laptop Sales dataset. By performing **Exploratory Data Analysis (EDA)**, key product attributes such as price, discount, ratings, and number of reviews were examined to understand their distributions, relationships, and overall influence on product performance.

A variety of visualization methods-including histograms, bar plots, count plots, and heatmaps-were employed to effectively uncover patterns, correlations, and variations within the data. Additionally, essential data cleaning and transformation steps, such as converting RAM and storage values into numeric form and extracting brand and OS information, significantly enhanced data consistency and analytical accuracy.

The visual analysis enabled the extraction of meaningful insights related to pricing structures, discount strategies, and customer engagement across different laptop brands and variants. These findings provide valuable business intelligence and establish a solid foundation for future work, such as **predictive modeling, demand forecasting, or recommendation systems**.

REFERENCES

1. Amazon India laptop product listings - Dataset source used for exploratory data analysis (EDA).
2. McKinney, W. (2017). *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and /Python*. O'Reilly Media.
3. Python Software Foundation. *Python Documentation*. <https://docs.python.org/>
4. Pandas Development Team. *Pandas Documentation*. <https://pandas.pydata.org/docs/>
5. Matplotlib Development Team. *Matplotlib Documentation*. <https://matplotlib.org/stable/contents.html>
6. Seaborn Development Team. *Seaborn Documentation*. <https://seaborn.pydata.org/>